Formal Contract for the Primary Effective-Service Allocation Task

Selective-Discreteness Downstream-Task Review — T6 Proposal

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Technical summary

The task allocates continuous effective signal service among four protected movement groups at one rotationally symmetric four-leg junction. It tests whether continuous traffic mass augmented by exact order and route/movement metadata yields more faithful and useful control sensitivities than homogeneous, movement-queue, and lane/storage-aware aggregate representations. The action is not claimed to be an executable phase schedule. Geometry and traffic variants are defined separately in `primary_green_allocation_suite.md`.

Notation convention. Fix one arbitrary model realization M before the following within-model definitions and derivations. To avoid carrying an unchanging superscript or subscript through every equation, write X, F, J, f, S, A, and B without M. The equations apply separately to every model realization. A model label is restored only where different models or a richer evaluator are explicitly compared.

1 Scenario and model contract

For the fixed model, scenario ξ, and service decision g, let

$$X_{0:K}(g;\xi) = \mathcal{S}(g;\xi)$$

denote the complete simulated trajectory. The scenario contains topology, link and lane geometry, fundamental-diagram parameters, phase–movement permissions, saturation flows, boundary demand, downstream supply, route table, fixed route shares, ordered route-mass realization, initial state, and (for an external evaluator) a fixed replication seed. These quantities are structural during differentiation.

Traffic quantity is continuous vehicle-equivalent mass. For every step k, total accounted mass satisfies

$$A_k = N_k^{\mathrm{src}} + N_k^{\mathrm{net}} + D_k,$$

where A_k is cumulative demand released up to step k, N_k^{src} is released mass still waiting in source queues, N_k^{net} is all mass physically present on modeled links and lane zones, and D_k is cumulative completed mass. In particular, N_k^{net} includes both moving and queued traffic; it is not a queue-length variable. No representation may drop mass to satisfy storage or service constraints.

The core scenario is invariant under a 90-degree rotation ρ: corresponding inbound/outbound geometry, traffic parameters, demand profiles, route shares, initial aggregate state, and downstream supply are identical. For V1 and V3, the ordered route-mass pattern is also copied by rotation. If P_ρ permutes the four service groups under that rotation, a conforming model must satisfy

$$f(P_\rho g;\rho\xi) = f(g;\xi), \qquad \nabla_g f(P_\rho g;\rho\xi) = P_\rho \nabla_g f(g;\xi).$$

This is a construction and implementation check. V4 may add explicitly labeled asymmetric held-out cases after the symmetric core is established.

2 Effective-service decision

Let $p \in \{1,\dots,P\}$ index fixed protected effective-service groups, with $P = 4$. Groups 1 and 2 serve north–south through/right and protected-left movements respectively; groups 3 and 4 serve the corresponding east–west movements. The physical allocation is

$$g_p \geq \underline{g}_p, \qquad \sum_{p=1}^{P} g_p = G, \qquad 0 < G \leq 1,$$

where g_p and G are dimensionless effective-service fractions. A differentiable default parameterization uses free logits $z \in \mathbb{R}^P$:

$$g_p(z) = \underline{g}_p + \left(G - \sum_{q=1}^{P} \underline{g}_q \right) \frac{e^{z_p}}{\sum_{q=1}^{P} e^{z_q}}.$$

The movement service-capacity rate is

$$c_m(g) = s_m \sum_{p=1}^{P} a_{mp} g_p,$$

where s_m has units veh/s and $a_{mp} \in \{0, 1\}$ is fixed phase–movement compatibility. This is a continuum capacity constraint. It does not specify green, yellow, or all-red intervals within a cycle. The four groups are therefore allocation categories, not an executable four-stage controller. Clearance time is neither a fifth group nor part of g; it belongs to a later timing-realization contract.

Physical comparisons should be reported in feasible tangent directions d satisfying $\sum_p d_p = 0$ and preserving lower bounds locally. Logit gradients are optimization coordinates, not the traffic-engineering result.

3 Objective and terminal treatment

The primary objective is clearance-complete total system time, including source queues:

$$J(g; \xi) = \Delta t \sum_{k=0}^{K-1} \left(N_k^{\mathrm{src}} + N_k^{\mathrm{net}} \right), \qquad [J] = \text{veh s}.$$

This is the discrete-time area under the number-in-system curve: every released unit of traffic contributes one unit of time at each step until it completes, whether it is waiting at a source, moving on a link, or queued. Therefore, when all released demand completes, the sum equals the total elapsed source-to-destination time of that traffic, up to the ordinary time-discretization convention. Summing only stopped or queued mass would instead define total waiting/queue time and is not the proposed objective.

Demand is released during a declared control/inflow window. The simulation then applies a fixed, model-common clearance policy until all scenario demand has completed. A run that fails to clear by a predeclared safety cap is a failed/censored evaluation, not a silently truncated low objective.

When every trip completes, free-flow-subtracted delay is reported as

$$D(g; \xi) = J(g; \xi) - J^{\mathrm{ff}}(\xi),$$

where J^{ff} is independent of g for a fixed scenario. Thus $\nabla_g D = \nabla_g J$, while delay has a more familiar traffic interpretation. Queue length, throughput, blocked mass, bay occupancy, and spillback duration are secondary mechanism diagnostics and do not replace the common primary objective.

4 Derivative contract

The colleague-facing notation is fixed as follows. The scalar function is the clearance-complete total system time produced by model M in fixed scenario ξ,

$$f(x; \xi) \equiv J(g; \xi),$$

and the scientific input is the physical phase-service allocation,

$$x \equiv g = (g_1, g_2, g_3, g_4)^\top.$$

Thus the reported scientific gradient is $df/dx = \nabla_g J$ with units veh s per unit service fraction. If an optimizer uses logits z, those are only internal coordinates and

$$\nabla_z f = \left(\frac{\partial g}{\partial z} \right)^\top \nabla_g J.$$

Here $z = (z_1, z_2, z_3, z_4)^\top \in \mathbb{R}^4$ is the dimensionless, unconstrained score vector introduced by the simplex parameterization in Section 2: $g = g(z)$. It has no direct traffic-engineering meaning. Because adding the same constant

to all softmax logits leaves g unchanged, an implementation may center z or fix one reference logit to remove that redundant coordinate. Gradients across models are compared in the common physical g coordinates or as feasible directional derivatives, never as raw logit gradients.

For a fixed scenario and a stable local event/active-set signature, the model derivative in feasible direction d is

$$\mathrm{D}J(g;\xi)[d] = \nabla_g J(g;\xi)^\top d.$$

Automatic differentiation is checked in double precision against the central directional difference

$$\widehat{\mathrm{D}}_h J[d] = \frac{J(g+hd;\xi) - J(g-hd;\xi)}{2h}.$$

A check is interpretable only when $g \pm hd$ are feasible and the compared trajectories retain the declared stable signature. Event-boundary cases are reported separately; they are not smoothed away or removed from forward evaluation.

For models M_i and a richer evaluator R, local decision usefulness is tested using identical feasible perturbations δd_i, selected without observing the evaluator outcome:

$$\Delta J_R^{(i)} = J_R(g+\delta d_i;\xi) - J_R(g;\xi).$$

Directional sign, rank, and realized improvement are distinct measurements. Internal gradient correctness does not imply $\Delta J_R^{(i)} < 0$.

5 Full gradient path from objective to control

Let X_k collect all continuous state of the fixed model at step k: aggregate link mass and cumulative counts, source mass, lane-zone occupancy, ordered ledger masses, and any continuous service bookkeeping used by that model. Structural topology, route identifiers, movement/lane labels, and the ordering skeleton belong to ξ rather than X_k. Write

$$X_{k+1} = F_k(X_k, g; \xi), \qquad f(g;\xi) = \sum_{k=0}^{K-1} \ell_k(X_k;\xi),$$

where $\ell_k = \Delta t(N_k^{\mathrm{src}} + N_k^{\mathrm{net}})$. With $S_k = \partial X_k/\partial g$ and an initial state independent of g, the forward sensitivity recursion is

$$S_0 = 0, \qquad S_{k+1} = A_k S_k + B_k,$$

$$A_k = \frac{\partial F_k}{\partial X_k}, \qquad B_k = \frac{\partial F_k}{\partial g}, \qquad \nabla_g f = \sum_{k=0}^{K-1} S_k^\top \nabla_{X_k} \ell_k.$$

Reverse-mode AD computes the same chain in adjoint form and is the intended implementation. The recursion defines what must be connected; it does not require forward-mode storage in production.

At traffic-process level, the chain is

$$g \longrightarrow c_m(g) \longrightarrow y_{m,k} \longrightarrow X_{k+1} \longrightarrow (N_{k+1}^{\mathrm{src}}, N_{k+1}^{\mathrm{net}}) \longrightarrow f,$$

where $y_{m,k}$ is realized movement discharge. V1–V3 insert additional state-dependent bounds and updates between movement capacity and future occupancy. Within a stable signature, active minima, capped transfers, prefix splits, and storage constraints use their ordinary branch derivatives. At ties or signature changes the classical derivative may not exist; no derivative through a branch choice, route label, or order permutation is invented.

6 Variant-specific gradient propagation

The variants share $f = J$ and $x = g$. What changes is the state transition F_k and therefore the path by which a service change affects future occupancy.

V0: movement-exclusive lanes, no bay

For each movement, realized discharge has the schematic form

$$y_{m,k} = \min\{D_{m,k},\ R_{m,k},\ c_m(g)\Delta t\},$$

where $D_{m,k}$ is available movement demand and $R_{m,k}$ is downstream receiving allowance, both in veh for the step. The direct path is $g \to c_m \to y_{m,k} \to X_{k+1} \to f$. Because each lane is movement-exclusive and full length, there is no cross-movement ordered-access derivative. V0 requires the macro, lane, and selective realizations to recover the same active-branch sensitivity.

V1: shared-lane order, no bay

The shared lane adds an eligible ordered-prefix bound $E_{m,k}(X_k; \xi_{\text{ord}})$:

$$y_{m,k} = \min\{E_{m,k},\ R_{m,k},\ c_m(g)\Delta t\}.$$

The ordering skeleton and movement labels ξ_{ord} are fixed. Gradients propagate through continuous prefix masses and partial removal: g changes discharge, which changes the residual head and eligible prefix at later steps, which changes future discharge and f. No gradient is taken with respect to swapping vehicles or changing labels. Aggregate-matched V1 pairs therefore have the same x but different fixed structural scenarios and separate values of df/dx.

V2: finite turning-bay storage, order homogenized

Let O_k^{bay} be bay occupancy, K^{bay} fixed storage, r_k^{bay} available bay receiving mass, b_k bay admission, and y_k^{bay} bay discharge. Schematically,

$$r_k^{\text{bay}} = \mathcal{R}(K^{\text{bay}} - O_k^{\text{bay}}), \quad b_k = \min\{D_k^{\text{LT}}, r_k^{\text{bay}}\}, \quad O_{k+1}^{\text{bay}} = O_k^{\text{bay}} + b_k - y_k^{\text{bay}}.$$

The added temporal path is $g \to y_k^{\text{bay}} \to O_{k+1}^{\text{bay}} \to r_{k+1}^{\text{bay}} \to b_{k+1}$, with a further path from rejected admission to feeder occupancy, upstream receiving restriction, source/network time, and f. Ordering is homogenized or deliberately nonbinding, so this derivative measures finite storage and spillback rather than exact access order.

V3: finite bay with ordered accessibility

V3 replaces unconstrained aggregate bay demand by an ordered-access operator

$$b_k = \mathcal{A}(L_k^{\text{ord}}, I_k^{\text{lane}}, r_k^{\text{bay}}),$$

where L_k^{ord} contains continuous ordered route masses and I_k^{lane} is continuous realized lane occupancy/access state under fixed permissions and order skeleton. The full added feedback path is

$$g \to y_k \to (O_{k+1}^{\text{bay}}, I_{k+1}^{\text{lane}}, L_{k+1}^{\text{ord}}) \to \mathcal{A}_{k+1} \to b_{k+1} \to y_{k+1} \to f.$$

Within a fixed scan/prefix/active-set signature, gradients propagate through continuous admitted, residual, split, and discharged masses. They do not propagate through route identity, movement labels, lane permissions, or a discrete permutation. V3 is the decisive test because the lane/storage aggregate baseline can share O^{bay} and r^{bay} but lacks the exact L^{ord}-conditioned accessibility path.

V4: held-out combined family

V4 introduces no new local traffic equation. For a fixed bank $\mathcal{B}$ of V1–V3 scenarios,

$$f(g;\mathcal{B}) = \frac{1}{|\mathcal{B}|}\sum_{\xi\in\mathcal{B}} f(g;\xi), \qquad \nabla_g f(g;\mathcal{B}) = \frac{1}{|\mathcal{B}|}\sum_{\xi\in\mathcal{B}} \nabla_g f(g;\xi).$$

Each scenario retains its own fixed structural order and active-set qualification. Development-bank gradients may drive optimization; held-out scenarios and richer-evaluator seeds are evaluation only and do not feed gradients back into the optimizer.

7 Aggregate-matched order pairs

An order pair (ξ^a, ξ^b) must match topology, geometry, demand time series, total approach mass, movement totals, nominal lane intent, initial aggregate occupancy, service decision, and downstream supply. Let Π_{agg} remove order/access detail. Then

$$\Pi_{\text{agg}}(\xi^a) = \Pi_{\text{agg}}(\xi^b), \qquad \xi^a_{\text{ord}} \neq \xi^b_{\text{ord}}.$$

The central selective-information test asks whether the richer reference exhibits a material contrast

$$J_R(g;\xi^a) - J_R(g;\xi^b) \quad \text{or} \quad \mathrm{D}J_R(g;\xi^a)[d] - \mathrm{D}J_R(g;\xi^b)[d],$$

and whether the selective-order model captures it more faithfully than aggregate alternatives. A pair with no richer-reference contrast is evidence that order is immaterial in that regime, not a failed dataset to discard.

8 Scenario-bank and optimization contract

For a fixed scenario bank $\mathcal{B}$, the sample-average task is

$$\bar{J}(g;\mathcal{B}) = \frac{1}{|\mathcal{B}|}\sum_{\xi\in\mathcal{B}} J(g;\xi).$$

Related members of an aggregate-matched pair remain in the same tuning or held-out split. Scenario identities, route/order realizations, and evaluator seeds are fixed before model comparison. Full optimization begins only after forward and derivative checks. Representation comparisons hold the action fixed; optimizer comparisons hold the model, feasible set, initialization policy, information, and evaluation budget fixed.

9 Result interpretation

Result class	Required interpretation
Macro equivalence	V0 representations agree within an approved numerical tolerance; failure blocks later attribution.
Storage sufficiency	The lane/storage aggregate model matches selective and richer results in V2; exact order is unnecessary for that regime.
Selective value	V3 richer-reference contrasts are captured more faithfully by the selective model, and its directions or actions transfer more usefully.
Partial value	Improvement occurs only in declared demand, storage, or ordering strata; claims remain conditional.
Negative transfer	Internal derivatives are correct but do not improve richer outcomes; differentiability is valid but downstream usefulness is unsupported.
Model failure	Conservation, service, feasibility, or derivative-reference checks fail; optimization evidence is inadmissible until corrected.

10 Decision status and boundary

The effective-service family and fixed structural differentiation boundary were accepted at T5. The equations above, clearance-complete objective, scenario matching, and evidence logic are proposed for T6 approval. Exact numeric geometry, demand, horizon, perturbation, tolerance, seed-count, and budget values remain defaults to calibrate and freeze in later implementation/experiment plans. Executable timing, offsets, cycle optimization, actuated control, endogenous route choice, and microscopic lane-changing dynamics remain deferred.